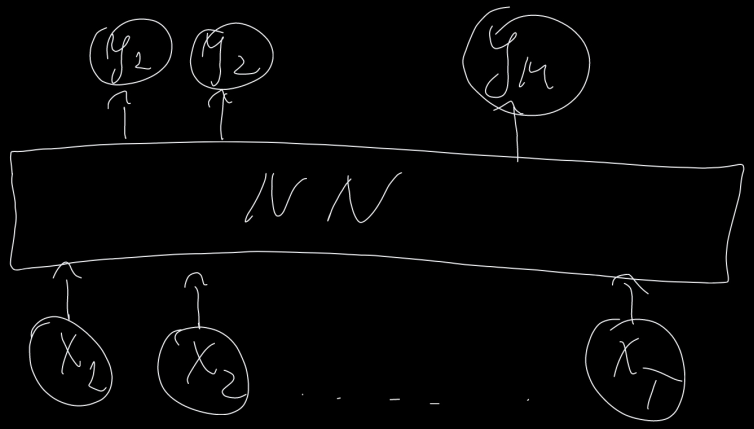
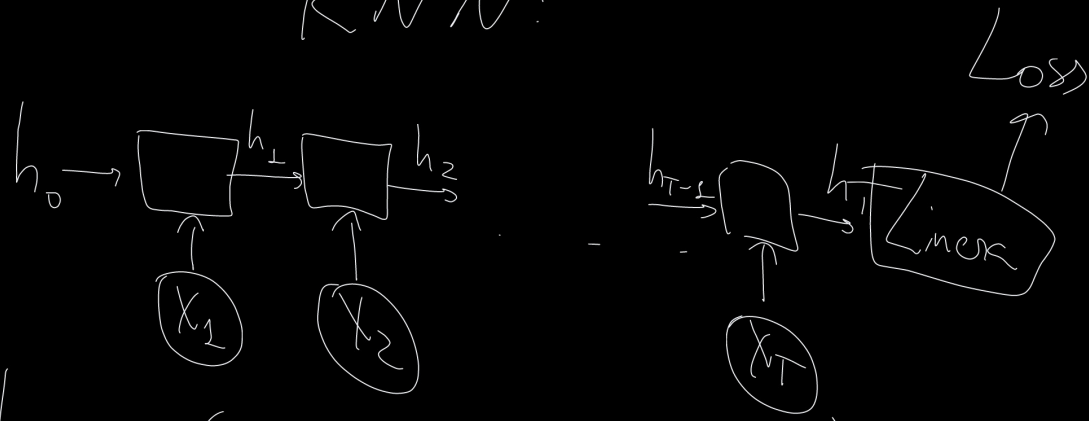
Recurrent NNs and Attention



$x_i \in \mathbb{R}^d$

RNN:

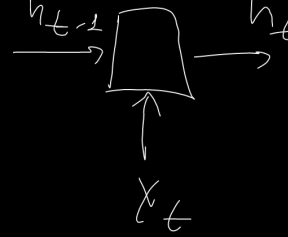


$$h_t = g(W_x x_t + W_h h_{t-1} + b)$$

$W_x \in \mathbb{R}^{\text{hid_dim} \times \text{inp_dim}}$

$W_h \in \mathbb{R}^{\text{hid_dim} \times \text{hid_dim}}$

$b \in \mathbb{R}^{\text{hid_dim}}$



Given: $\nabla_{h_t} L$ Find: $\nabla_{h_{t-1}} L$

$$h_t = g(\underbrace{W_x x_t + W_h \cdot h_{t-1} + b}_{z_t})$$

$$dL = \nabla_{h_t} L^T dh_t = \nabla_{h_t} L^T \frac{\partial g}{\partial z_t} dz_t =$$

$$= \underbrace{\nabla_{h_t} L^T \frac{\partial g}{\partial z_t}}_{\nabla_{h_t} L^T} W_h \cdot dh_{t-1} \Rightarrow \nabla_{h_{t-1}} L = W_h^T \left(\frac{\partial g}{\partial z_t}\right)^T \nabla_{h_t} L =$$

$$\|\nabla_{h_{t-1}} L\| \leq \left(\prod_{k=t}^{T} \|W_h\| \left\|\frac{\partial g}{\partial z_k}\right\|\right) \|\nabla_{h_T} L\| = \left[\prod_{k=t}^{T} W_h^T \left(\frac{\partial g}{\partial z_k}\right)\right] \nabla_{h_T} L$$

Exploding gradients: gradient clipping

$$g = \begin{cases} N \cdot \frac{g}{\|g\|}, & \text{if } \|g\| > N \\ g, & \text{otherwise} \end{cases} \qquad N = 1,5$$

$g = \nabla L$

$$\|g\| = \sqrt{\sum_{i=1}^{d} g_i^2}$$

$g_i \approx 10^{-4}$, $g_i^2 \approx 10^{-8}$

$d = 10^6$

$\sum g_i^2 \approx 10^{-2}$

$\sqrt{\sum g_i^2} \approx 10^{-1}$

Vanishing gradients:

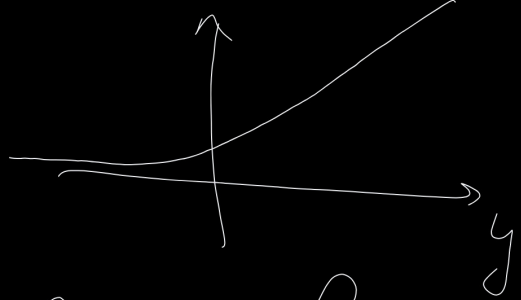
1) Use ReLu, Leaky ReLu
2) Use orthogonal W_h:
   - init. W_h with orthogonal
   - add regularizer: $\lambda \| W_h^T W_h - I \|_F^2$
   - orth. reparameterization
3) Use LSTM, GRU units

$$f(x) \to \min_{x \in C}$$

$$x = h(y), \quad y \in R^d, \quad x \in C$$

$$f(x) = f(h(y)) \to \min_y$$

$x \geq 0$ $\quad x = \log(1 + \exp(y))$ $\quad h(y) = \exp(y)$



$X = X^T \succeq 0$

$a \leq x \leq b$ $\quad x = a + \sigma(y)(b - a)$ $\quad X = YY^T$

$X^T X = I \Rightarrow X = \text{expm}(Y - Y^T)$ - orth.

LSTM (Long Short Term Memory)

C_t – long term memory

h_t – short term memory

$$C_t = f_t \odot C_{t-1} + i_t \odot g_t$$

$$h_t = o_t \odot \tanh(C_t)$$

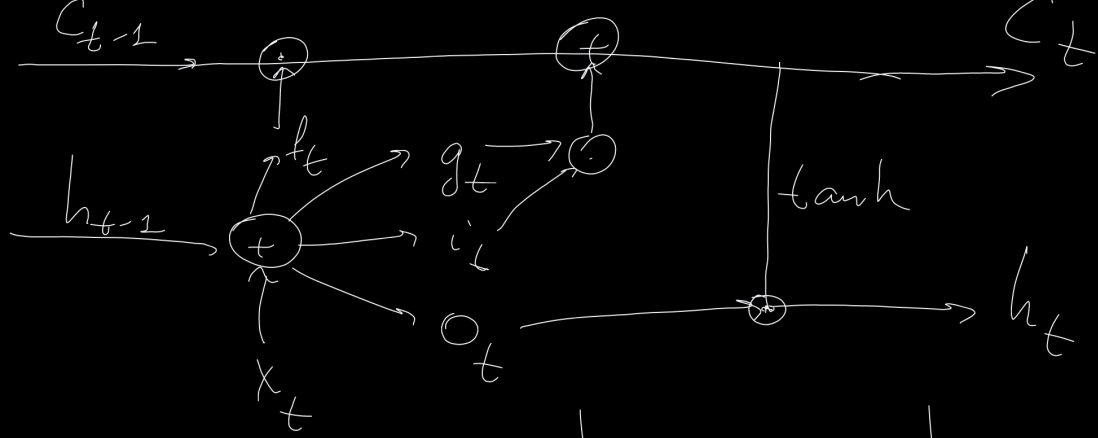
$$g_t = g\left(W_x^g \cdot x_t + W_h^g \cdot h_{t-1} + b^g\right)$$

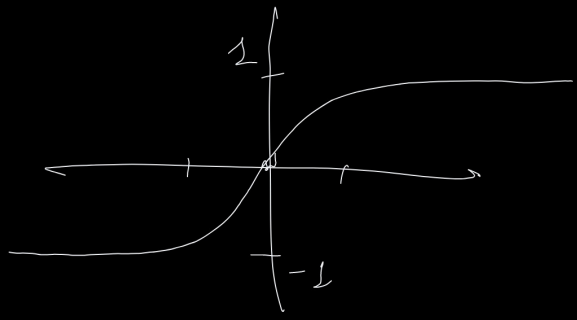
$$i_t = \sigma\left(W_x^i \cdot x_t + W_h^i \cdot h_{t-1} + b^i\right) \in (0,1)$$

$$o_t = \sigma\left(W_x^o \cdot x_t + W_h^o \cdot h_{t-1} + b^o\right) \in (0,1)$$

$$f_t = \sigma\left(W_x^f \cdot x_t + W_h^f \cdot h_{t-1} + b^f\right) \in (0,1)$$

Init: $b^f >> 1$





$$C_t = f_t \odot C_{t-1} + i_t \odot g_t$$

$$f_t \approx 1 \qquad \frac{\partial C_t}{\partial C_{t-1}} = I$$

input	output	forget	Behavior of LSTM cell
+	+	−	Classic RNN
+	−	+	Store inform. into memory
−	+	+	Load inform. from memory
−	−	+	Skipping current input

GRU (Gated Recurrent Unit)

$$r_t = \sigma\left(W_x^r x_t + W_h^r h_{t-1} + b^r\right) \quad \text{– reset gate}$$

$$u_t = \sigma(\ldots) \quad \text{– update gate}$$

$$g_t = g\left(W_x^g x_t + W_h^g\left(h_{t-1} \odot r_t\right) + b^g\right)$$

$$h_t = u_t \odot h_{t-1} + (1 - u_t) \odot g_t$$

Init: $b^u >> 1$

$$net = nn.LSTM(input_dim, hidden_dim, num_layers, dropout=0.2, batch_first=True, \dots)$$

$$out, (h_T, c_T) = net(input, (h_0, c_0))$$

$input \in \mathbb{R}^{B \times T \times input_dim}$

$h_0, c_0 \in \mathbb{R}^{L \times B \times H}$

$out \in \mathbb{R}^{B \times T \times H}$

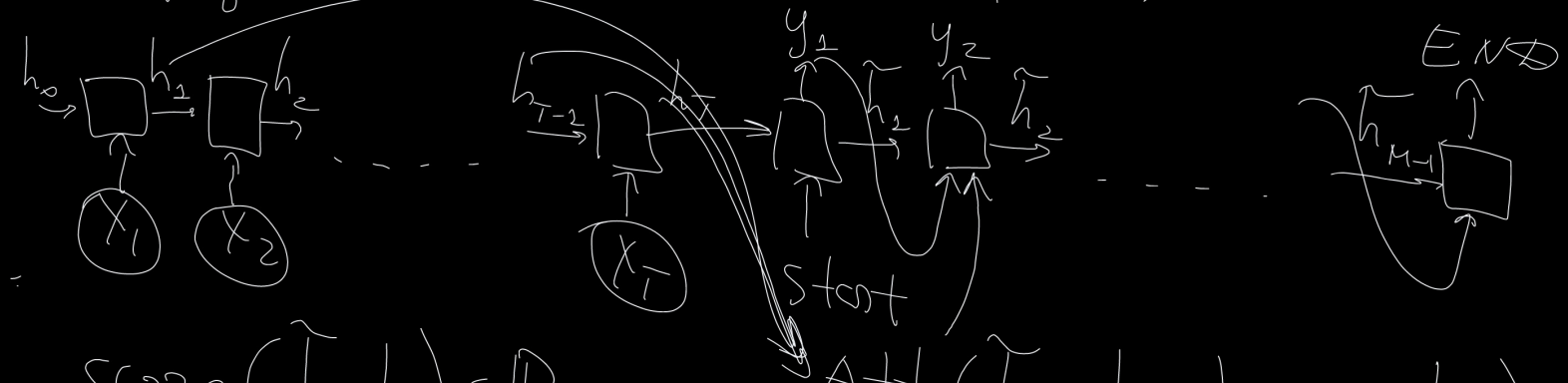
$h_T, c_T \in \mathbb{R}^{L \times B \times H}$

```
LSTMCell
for t=0,1,...,T:
    h,c = LSTMCell(x[:,t,:], h, c)

net = CustomLSTM(...)
net = torch.compile(net)
```

Attention mechanism

Copy task: acdeaad → acdeaadacdeaad



$score(\tilde{h}, h) \in \mathbb{R}$

$Att(\tilde{h}_1, h_1, h_2, \ldots, h_T)$

$$\alpha_{ti} = \frac{\exp(score(\tilde{h}_t, h_i))}{\sum_{j=1}^{T} \exp(score(\tilde{h}_t, h_j))}, \quad i = 1 \ldots T \qquad Att_t = \sum_{i=1}^{T} \alpha_{ti} h_i$$

$score(\tilde{h}, h)$:

① $\tilde{h}^T h$

② $\frac{\tilde{h}^T h}{\sqrt{dim(\tilde{h})}}$

③ $w^T \tanh(W_1 \tilde{h} + W_2 h + b)$

$h_i \sim N(0, 1)$

$\mathbb{E}\, \tilde{h}^T h = \mathbb{E} \sum \tilde{h}_i h_i =$

$= \sum \mathbb{E}\tilde{h}_i \cdot \mathbb{E} h_i = 0$

$Var(\tilde{h}^T h) = \sum_{i=1}^{d} Var(\tilde{h}_i) Var(h_i) = d$

$1 \quad 1$